Costs and the Firm's Supply Curve

EC306: Intermediate Microeconomics — Day 5

Core reading: Perloff (2020), Ch. 7 and §8.1–8.4

The final step: from technology to supply

Where we are

Day 4 gave us $q = f(L, K)$ — pure engineering. It says nothing about *which* input mix the firm uses. That requires **prices**.

1. Add input prices $(w, r) \Rightarrow$ minimise cost $\Rightarrow$ the **cost function** $C(q)$.
2. Slice $C(q)$ into the family of short-run cost curves.
3. Add the *output* price p and profit maximisation $\Rightarrow$ the **supply curve**.

By the end of today the demand–supply diagram is fully derived from first principles.

Cost minimisation = the firm's tangency problem

$$\min_{L,K} \; wL + rK \quad \text{s.t.} \quad f(L,K) = q.$$

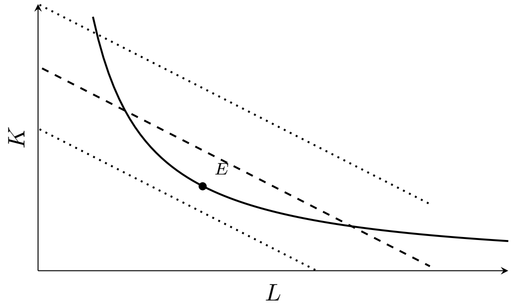
Isocost lines $wL + rK = C$ have slope $-w/r$ — the firm's "budget lines". Tangency to the isoquant:

$$\boxed{\text{MRTS} = \frac{\text{MP}_L}{\text{MP}_K} = \frac{w}{r}}$$

The same equal-bang-for-buck logic

$\dfrac{\text{MP}_L}{w} = \dfrac{\text{MP}_K}{r}$: the last pound spent on labour must add the same output as the last pound on capital. If not, re-mix inputs and produce the same q for less.

Cost minimisation, drawn



The cost-minimising input mix for output q is where the isoquant touches the *lowest* attainable isocost line, E. Lower (dotted) lines can't reach q; higher ones waste money.

Worked example — a cost function from Cobb–Douglas

$q = L^{1/2}K^{1/2}$, input prices w, r. Tangency: $\dfrac{\mathrm{MP}_L}{\mathrm{MP}_K} = \dfrac{K}{L} = \dfrac{w}{r} \Rightarrow K = \dfrac{w}{r}L$. Substitute into $q = L^{1/2}K^{1/2} = L\sqrt{w/r}$, so the conditional demands are

$$L^\star = q\sqrt{r/w}, \qquad K^\star = q\sqrt{w/r}.$$

$$\boxed{C(q) = wL^\star + rK^\star = 2q\sqrt{wr}}$$

Read it

Cost is *linear* in q here (constant returns to scale $\Rightarrow$ constant marginal cost $2\sqrt{wr}$). The $\sqrt{wr}$ term shows costs rise with either input price, dampened because the firm re-mixes toward whichever input got relatively cheaper.

Short-run cost concepts

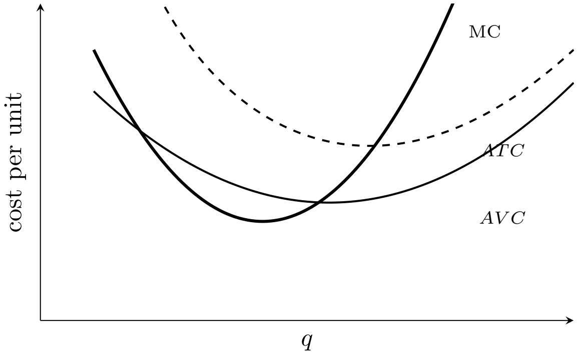
With $K = \bar{K}$ fixed, total cost splits into fixed and variable:

$$C(q) = \underbrace{F}_{\text{fixed}} + \underbrace{V(q)}_{\text{variable}} .$$

- $AFC = F/q$ (always falling)
- $AVC = V(q)/q$
- $ATC = C(q)/q = AFC + AVC$
- $\text{MC} = \dfrac{\mathrm{d}C}{\mathrm{d}q} = \dfrac{\mathrm{d}V}{\mathrm{d}q}$
- (MC ignores F — fixed cost is sunk in the SR)

The single most important relationship is between MC, AVC, and ATC — next slide.

The cost-curve relationships



MC **cuts both AVC and ATC at their minimum points.** The AVC–ATC gap is AFC, which shrinks as q grows (the curves converge). Memorise this picture — it *is* the supply curve, as we now show.

Long-run cost: the envelope

In the LR all inputs flex, so the firm picks the best plant size for each q. The **LR average cost** is the lower envelope of all the SR ATC curves.

Shape and its source

- Falling LRAC = *economies of scale* $\leftrightarrow$ increasing returns to scale.
- Flat = constant returns. Rising = *diseconomies* $\leftrightarrow$ decreasing returns.

Tie back to Day 4

The U-shaped LRAC is the cost-side image of returns to scale: scale economies first (spreading fixed costs, specialisation), then diseconomies (management strain). The minimum of LRAC is "minimum efficient scale" — a key number in any market-structure analysis.

Profit maximisation gives $p = \mathrm{MC}$

A price-taking firm maximises $\pi = pq - C(q)$:

$$\frac{\mathrm{d}\pi}{\mathrm{d}q} = p - \mathrm{MC}(q) = 0 \quad \Longrightarrow \quad \boxed{p = \mathrm{MC}(q)}$$

plus the second-order condition that MC is rising.

But not all of MC is the supply curve

$p = \mathrm{MC}$ tells the firm the *best* output *if* it produces at all. Whether it should produce, or shut down, is a separate test — and it differs between short and long run. That test is what carves the supply curve out of the MC curve.

The shutdown rule (short run)

In the SR, fixed cost F is sunk — paid whether or not you produce. So produce iff revenue covers *variable* cost:

$$pq \geq V(q) \quad \Longleftrightarrow \quad p \geq AVC.$$

Short-run supply

$$\boxed{\text{SR supply} = \text{MC}(q) \text{ for } p \geq \min AVC; \quad q = 0 \text{ otherwise.}}$$

Why airlines fly half-empty flights at a loss

If a scheduled flight covers fuel, crew, and handling (its variable costs) but not its share of aircraft leasing (fixed/sunk in the SR), flying still beats cancelling — it contributes to fixed costs already committed. The shutdown rule is literally how that call is made.

The exit rule (long run)

In the LR nothing is sunk — the firm can shed F by exiting. So it produces iff it covers *all* costs:

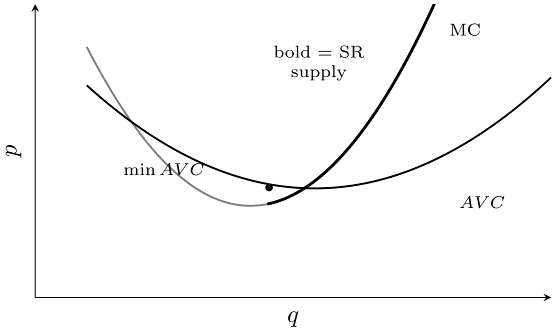
$$p \geq ATC \text{ (equivalently } \min \; LRAC).$$

Long-run supply (firm)

$$\boxed{\text{LR supply} = \text{MC}(q) \text{ for } p \geq \min ATC; \quad \text{exit otherwise.}}$$

SR threshold = $\min AVC$; LR threshold = $\min ATC$. The only difference is whether F is escapable. That single distinction is the whole of the SR/LR supply story.

Supply = the MC curve above the relevant threshold



Below min AVC the firm supplies zero; above it, supply is the rising MC curve. *This object, summed across firms, is the market supply curve* — the line we drew on day one without justification.

Your turn (1) — in-lecture

Your turn

A firm has $C(q) = q^2 + 4q + 25$.

1. Find MC, AVC, ATC.
2. Find the shutdown price (min AVC) and the break-even price (min ATC).
3. Write the short-run supply curve $q(p)$.

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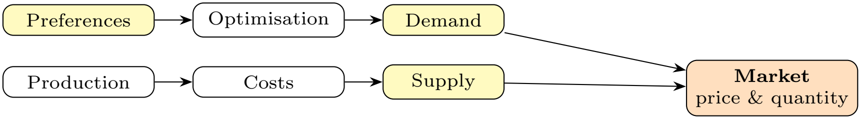
Your turn (2) — in-lecture

Your turn

Using the firm above, at $p = 10$:

1. How much does it supply?
2. Is it making a profit? Should it produce in the short run? In the long run?

The five days, assembled



What you built

You derived both blades of the Marshallian scissors from primitives — preferences and technology — with the same optimisation logic on each side. That is the core analytical skill of intermediate microeconomics. Everything else (market structure, welfare, game theory) builds on exactly this.